Breasts and Nipples Aplasia or Hypoplasia 2 (BNAH2) — Comprehensive Disease Report

MONDO: MONDO:0014450 · **OMIM:** 616001 · **Gene:** *PTPRF* (LAR) · **Inheritance:** Autosomal recessive · **Category:** Mendelian, ultra-rare

Summary

Breasts and Nipples Aplasia or Hypoplasia 2 (BNAH2) is an ultra-rare, congenital, autosomal recessive disorder of nipple-areola and breast development caused by **biallelic loss-of-function variants in *PTPRF***, the gene encoding the LAR (leukocyte common antigen-related) receptor-type protein tyrosine phosphatase. The condition was molecularly delineated in a single consanguineous Israeli Arab family in which three affected children across two branches of the pedigree carried a homozygous frameshift mutation in *PTPRF*, identified by SNP-array homozygosity mapping and candidate gene sequencing (PMID: 24781087). The phenotype is essentially isolated — a spectrum ranging from unilateral hypothelia (hypoplastic nipple) to bilateral athelia (absent nipple) with breast aplasia/hypoplasia, accompanied by a characteristic eyebrow shape, and no other consistently associated congenital or ectodermal anomalies.

Mechanistically, *PTPRF*/LAR is a transmembrane receptor tyrosine phosphatase that localizes to adherens junctions and participates in the regulation of epithelial cell–cell contacts, peptide growth-factor signaling, and the canonical Wnt pathway. Loss of LAR phosphatase activity is inferred to disrupt epithelial adhesion and signaling during development of the mammary ridge (milk line), producing the nipple and breast defect. This human causality is strongly corroborated by the mouse model: *Ptprf*(LAR)-deficient female mice develop normally but display impaired terminal differentiation of mammary alveoli at late pregnancy and a resulting lactation defect, demonstrating a conserved, essential role for LAR in mammary gland development and function (PMID: 9245518).

Clinically, BNAH2 carries an **excellent prognosis** with normal lifespan and no systemic involvement; the primary impacts are cosmetic, functional (inability to breastfeed in affected females), and psychosocial. There is **no disease-modifying or pharmacologic therapy**; management is reconstructive (breast and nipple-areola reconstruction via prosthetic, autologous, or fat-grafting techniques) and supportive. Because disease arises almost exclusively from shared ancestral (founder) homozygosity in consanguineous unions, prevention centers on **genetic counseling, carrier/cascade testing, and prenatal or preimplantation genetic options**. The disease is distinguished from the far more common *syndromic* causes of athelia/amastia — ulnar-mammary syndrome (*TBX3*), TP63-related ectodermal dysplasias, scalp-ear-nipple (Finlay-Marks) syndrome (*KCTD1*), and KMT2D-related disorder — by its isolated nature and recessive inheritance.

Key Findings

Finding 1 — BNAH2 is isolated congenital athelia/breast hypoplasia caused by biallelic *PTPRF* loss-of-function

The disease entity **MONDO:0014450** resolves to "breasts and/or nipples, aplasia or hypoplasia of, 2" (BNAH2), with cross-references to **OMIM:616001**, **GARD:0016044**, and **MedGen/UMLS C4014918**. The molecular cause was established by Borck and colleagues in 2014, who studied three children (one girl, two boys) from two branches of an extended consanguineous Israeli Arab family. Using **homozygosity mapping after SNP-array genotyping followed by candidate-gene sequencing**, they identified a **homozygous frameshift mutation in *PTPRF*** as the likely cause of the nipple anomalies.

> *"Using homozygosity mapping after single nucleotide polymorphism (SNP) array genotyping and candidate gene sequencing we identified a homozygous frameshift mutation in PTPRF as the likely cause of nipple anomalies in this family."* — PMID: 24781087

The affected individuals "presented with a spectrum of nipple anomalies ranging from unilateral hypothelia to bilateral athelia but no other consistently associated anomalies except a characteristic eyebrow shape," establishing both the **phenotypic spectrum** and the

essentially isolated (non-syndromic) nature of the condition. This finding anchors the entire disease entry: BNAH2 is a Mendelian, recessive, developmental defect of the nipple-areola complex and breast.

Finding 2 — Mechanism: *PTPRF*/LAR regulates nipple/mammary epithelial development via adherens junctions and Wnt signaling

PTPRF encodes the **LAR receptor-type protein tyrosine phosphatase**, which localizes to adherens junctions and regulates epithelial cell–cell contacts, peptide growth-factor signaling, and the canonical Wnt pathway.

> *"PTPRF encodes a receptor-type protein phosphatase that localizes to adherens junctions and may be involved in the regulation of epithelial cell-cell contacts, peptide growth factor signaling, and the canonical Wnt pathway."* — PMID: 24781087

The protein architecture comprises an extracellular domain of Ig-like plus fibronectin type-III repeats (cell-adhesion-molecule-like) and two intracellular PTP (protein tyrosine phosphatase) domains (PMID: 9245518). Population-genetic constraint data (gnomAD, GRCh38) indicate that *PTPRF* is **strongly intolerant to heterozygous loss-of-function**: pLI ≈ 1.0, LOEUF (observed/expected LoF upper bound) ≈ 0.36, LoF observed/expected ≈ 0.29, LoF-Z = 8.5, missense-Z = 4.6. This constraint is notable because BNAH2 is recessive — heterozygous carriers are unaffected for the nipple phenotype, yet complete LoF is apparently under strong purifying selection, implying essential biological roles beyond the breast.

Finding 3 — Mouse model: *Ptprf*(LAR)-deficient females show impaired alveolar differentiation and a lactation defect

Schaapveld et al. (1997) generated mice lacking both LAR phosphatase domains. **LAR$^{-/-}$ mice develop and grow normally with no appreciable histological abnormalities**, but pups of LAR$^{-/-}$ females showed abnormal neonatal death. The mammary glands of LAR$^{-/-}$ females were **incapable of delivering milk** owing to impaired terminal differentiation of alveoli at late pregnancy; the glands failed to switch to a lactational state and underwent rapid postpartum involution. Wild-type LAR expression is pregnancy-regulated, peaking around day 16 of gestation.

> *"Mammary glands of LAR-/- females were incapable of delivering milk due to an impaired terminal differentiation of alveoli at late pregnancy."* — PMID: 9245518

This model provides strong cross-species corroboration that *PTPRF*/LAR is required for mammary gland differentiation and function, directly supporting the inferred human causal mechanism. (Note the species difference: the mouse phenotype is predominantly a functional lactation defect, whereas the human phenotype is a structural nipple/breast developmental defect — see Limitations.)

Finding 4 — Phenotype spectrum: congenital, bilateral-or-asymmetric athelia/hypothelia and breast aplasia/hypoplasia with a characteristic eyebrow shape

Borck et al. describe three affected children with a spectrum from unilateral hypothelia to bilateral athelia and absence of the nipple-areola complex, plus a characteristic eyebrow shape, with **no other consistently associated congenital or ectodermal anomalies**. The defect is present from birth (congenital) and non-progressive.

HPO term mapping (verified via OLS/HP):

Phenotype	HPO term
Absent nipple / athelia	HP:0002561
Aplasia/Hypoplasia of the nipples	HP:0006709
Hypoplastic nipples	HP:0002557
Hypoplastic areola	HP:0100853
Breast aplasia	HP:0100783
Aplasia/Hypoplasia of the breasts	HP:0010311
Breast hypoplasia	HP:0003187
Unilateral breast hypoplasia	HP:0012813
Bilateral breast hypoplasia	HP:0012814
Abnormal eyebrow morphology	HP:0000534

> *"who presented with a spectrum of nipple anomalies ranging from unilateral hypothelia to bilateral athelia but no other consistently associated anomalies except a characteristic eyebrow shape"* — PMID: 24781087

Laterality is variable (unilateral to bilateral; asymmetric expression documented within the same family), indicating **variable expressivity** despite high penetrance in homozygotes.

Finding 5 — Differential diagnosis: BNAH2 (*PTPRF*) vs. syndromic causes of athelia/amastia

Athelia/amastia is usually *syndromic*. Distinguishing BNAH2 (isolated, autosomal recessive, *PTPRF*) from the major syndromic differentials is clinically critical:

Condition	Gene / Inheritance	Distinguishing features (absent in BNAH2)
Ulnar-mammary syndrome (UMS)	*TBX3* haploinsufficiency / AD	Ulnar ray defects, apocrine/genital anomalies, delayed puberty/hypogonadism, pituitary hypoplasia
TP63-related disorders (LMS/EEC/AEC)	*TP63* / AD	Ectrodactyly/split-hand-foot, cleft lip/palate, lacrimal duct stenosis, nail/dental anomalies, ectodermal dysplasia
Scalp-ear-nipple (Finlay-Marks) syndrome	*KCTD1* / AD	Aplasia cutis of scalp, prominent/malformed ears; recurrent p.(Ala30Glu), p.(Pro31Leu)
KMT2D missense (non-Kabuki)	*KMT2D* / de novo	Choanal atresia, hypoparathyroidism, short stature, interstitial lung disease
BNAH2	***PTPRF*** / **AR**	**Isolated nipple/breast defect + characteristic eyebrow; none of the above**

> *"Ulnar-mammary syndrome (UMS) is characterized by ulnar defects, and nipple or apocrine gland hypoplasia, caused by TBX3 haploinsufficiency."* — PMID: 30550377
>
> *"limb mammary syndrome (LMS), an autosomal dominant inherited disorder characterized by ectrodactyly, hypoplasia of mammary-gland and nipple, lacrimal duct stenosis, nail dysplasia, dental anomalies, cleft palate and/or cleft lip"* — PMID: 32067224

The KMT2D phenotypic expansion is a particularly important mimic because it includes athelia: *"including athelia (absent nipples), choanal atresia, hypoparathyroidism, delayed or absent pubertal development, and extreme short stature"* (PMID: 32083401). BNAH2 is distinguished by the **absence** of limb, ectodermal, scalp, apocrine, endocrine, and facial-dysmorphism features, and by its recessive inheritance in a consanguineous setting.

Finding 6 — Management is reconstructive/supportive; no disease-modifying therapy

No gene-directed or pharmacologic therapy exists for *PTPRF*-related athelia. Management is **surgical/cosmetic and psychosocial**. A systematic review of congenital breast anomaly reconstruction (Dreifuss et al., 2014) quantified the modalities used across 54 individual reconstructions:

Reconstruction modality	Patients	Percentage
Prosthetic (implant)	19	35.2%
Autologous	19	35.2%
Autologous + prosthesis	15	27.8%

> *"Prosthetic and autologous reconstructions were equally represented (19 patients each, 35.2%; Table 2). Autologous reconstruction with prosthesis was slightly less common (15 patients, 27.8%)."* — PMID: 23896162

Mean age at reconstruction was 27.6 years (range 13–54); the nipple-areola complex was addressed in 36 cases, with the majority (66.7%) not reconstructed. **Fat grafting** is a documented option for congenital amastia (PMID: 33830564). Timing considerations generally defer reconstruction until breast maturity/adolescence. Relevant NCIT clinical-intervention concepts include breast reconstruction, breast implant, autologous tissue reconstruction, and autologous fat transfer/grafting.

Finding 7 — Anatomy, cell types, and molecular annotations (UBERON/CL/GO/ortholog terms)

Affected anatomical structures (congenital developmental defect of the milk line / mammary ridge):

Structure	UBERON term
Nipple	UBERON:0002030
Areola	UBERON:0002032
Breast	UBERON:0000310
Mammary gland	UBERON:0001911

Tissue/cell level: ectoderm-derived mammary/skin epithelium. Relevant Cell Ontology terms (corrected — see Finding 9): epithelial cell (CL:0000066), mammary gland epithelial cell (CL:0002327), luminal epithelial cell of mammary gland (CL:0002326), basal-myoepithelial cell of mammary gland (CL:0002324), lactocyte (CL:0002325), keratinocyte (CL:0000312).

PTPRF protein (UniProt **P10586**, "Receptor-type tyrosine-protein phosphatase F"/LAR; gene MIM 179590; Entrez 5792) is a plasma-membrane receptor (GO:0005886) localizing to adherens junctions.

> *"PTPRF encodes a receptor-type protein phosphatase that localizes to adherens junctions"* — PMID: 24781087

GO annotations: molecular function — protein tyrosine phosphatase activity (GO:0004725), transmembrane receptor PTP activity (GO:0005001), cell adhesion molecule binding (GO:0050839); biological process — cell adhesion (GO:0007155), cell migration (GO:0016477), peptidyl-tyrosine dephosphorylation (GO:0035335), cell-surface receptor PTP signaling (GO:0007185).

Orthologs: mouse *Ptprf* (Entrez 19268), rat (360406), dog (475389), cow (512072), chicken (424568), zebrafish (140819).

Finding 8 — Epidemiology & inheritance: ultra-rare autosomal recessive, single-family, consanguinity-driven

BNAH2 (OMIM 616001) is defined molecularly from a single consanguineous Israeli Arab family with three affected children across two branches; it was identified by homozygosity mapping, consistent with **autosomal recessive inheritance from a shared ancestral (founder) *PTPRF* allele**.

> *"we report on three children from two branches of an extended consanguineous Israeli Arab family"* — PMID: 24781087

No prevalence or incidence figures exist; the disease is ultra-rare (<1/1,000,000; Orphanet "unknown prevalence" class). gnomAD shows that biallelic *PTPRF* loss-of-function is essentially absent in the general population (pLI ≈ 1.0; very few LoF carriers), so population carrier frequency is extremely low and disease occurs almost exclusively in consanguineous unions. **Both sexes are affected** (one girl and two boys in the index family). Penetrance of the nipple phenotype appears high in homozygotes, with **variable expressivity** (unilateral hypothelia to bilateral athelia). Consanguinity is the key modifiable risk factor.

Finding 9 — Ontology QC corrections and confirmed cross-references

OLS/CL verification corrected the mammary-epithelial Cell Ontology assignments used in the final annotations: the correct terms are **CL:0002327** (mammary gland epithelial cell), **CL:0002326** (luminal epithelial cell of mammary gland), **CL:0002324** (basal-myoepithelial cell of mammary gland), and **CL:0002325** (lactocyte). Importantly, CL:0002325 is *lactocyte* — **not** "basal cell of mammary gland" as mis-stated in an earlier draft; the basal/myoepithelial term is CL:0002324. Keratinocyte = CL:0000312; generic epithelial cell = CL:0000066. The MONDO:0014450 cross-references were re-confirmed: **OMIM:616001, GARD:0016044, MedGen/UMLS C4014918** (MedGen CUI 863355).

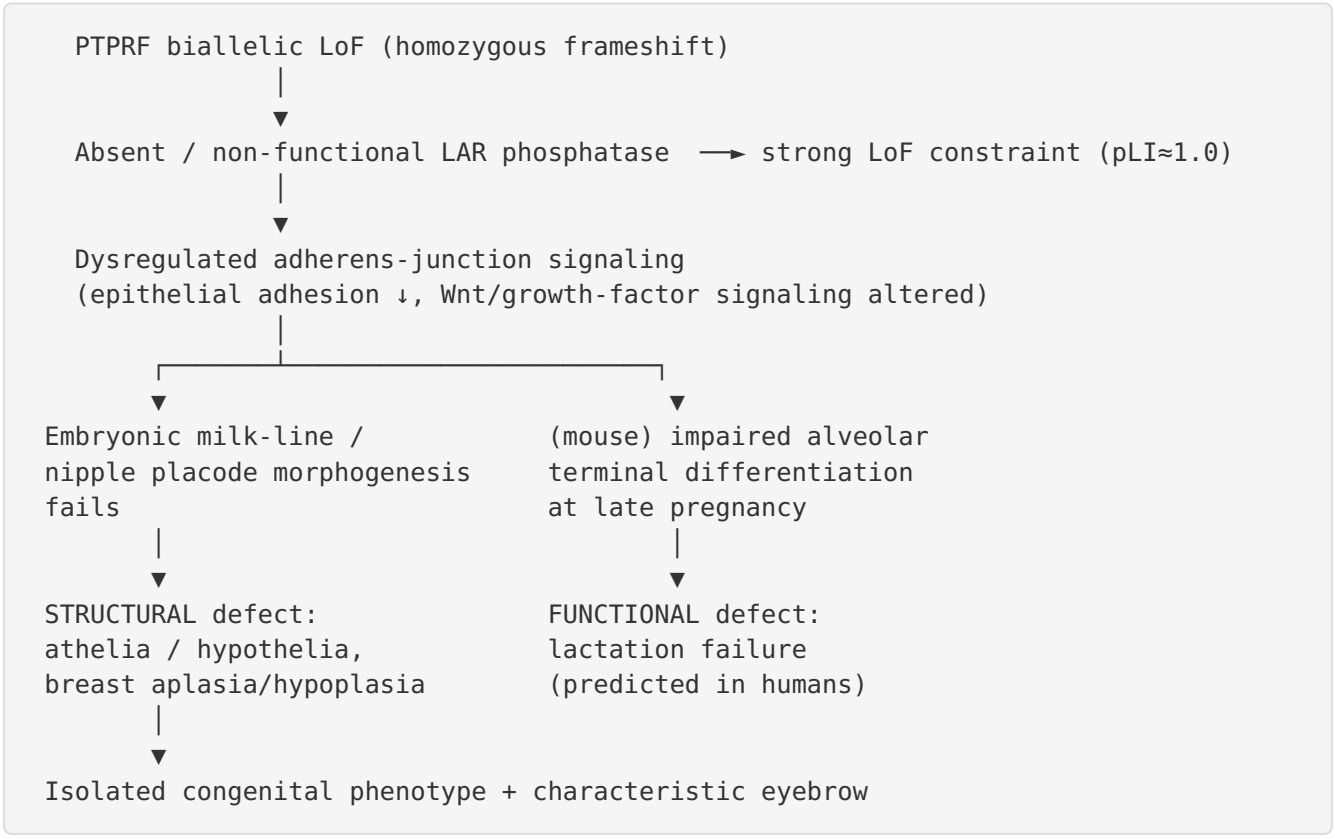
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic frameshift (loss-of-function) mutation in *PTPRF*** (homozygous, inherited from a shared ancestral allele in a consanguineous union) **leads to** absent or non-functional LAR receptor-type protein tyrosine phosphatase. *(Demonstrated — Borck 2014.)*
2. **Loss of LAR phosphatase activity at adherens junctions results in** dysregulated dephosphorylation of LAR substrates and impaired control of epithelial cell–cell contacts. *(Inferred from LAR molecular function.)*
3. **Impaired epithelial adhesion plus altered peptide growth-factor and canonical Wnt signaling disrupts** the coordinated proliferation, migration, and placode/bud formation of

mammary-ridge (milk-line) ectodermal epithelium during embryonic development. *(Inferred; Wnt/adhesion roles documented, developmental timing in humans not directly demonstrated.)*

4. **Failure of nipple placode / mammary bud morphogenesis produces** the structural defect: absent or hypoplastic nipple-areola complex and underdeveloped/absent breast parenchyma. *(Demonstrated phenotype; mechanistic link inferred.)*
5. **Branch (postnatal/functional):** In the analogous mouse model, surviving LAR-deficient epithelium **fails** terminal alveolar differentiation at late pregnancy, **causing** a lactation defect. *(Demonstrated in mouse; predicts impaired lactation in affected human females who retain some breast tissue.)*
6. **Clinical manifestation:** congenital athelia/hypothelia + breast aplasia/hypoplasia with a characteristic eyebrow shape, isolated and non-progressive. *(Demonstrated.)*



Upstream vs downstream: The *PTPRF* mutation and loss of LAR phosphatase activity are the upstream initiating events; adhesion/Wnt signaling disruption is intermediate; the structural nipple/breast defect is the downstream clinical endpoint. **Cell types involved:** ectoderm-derived mammary/skin epithelial cells (basal-myoepithelial CL:0002324, luminal CL:0002326, mammary epithelial CL:0002327, keratinocyte CL:0000312). **Key biological processes (GO):**

cell adhesion (GO:0007155), peptidyl-tyrosine dephosphorylation (GO:0035335), cell-surface receptor PTP signaling (GO:0007185), cell migration (GO:0016477), all converging on mammary gland morphogenesis.

The coherence of this model rests on three independent pillars: (1) human genetic evidence (homozygous LoF segregating with phenotype in a consanguineous pedigree), (2) molecular plausibility (LAR's documented adherens-junction/Wnt roles), and (3) an animal model demonstrating an essential mammary role for the orthologous gene.

Evidence Base

PMID	Title (abbrev.)	Evidence type	How it supports / challenges the findings
24781087	*Homozygous truncating PTPRF mutation causes athelia*	Human clinical + genetics	**Foundational.** Establishes *PTPRF* biallelic LoF as the cause (F001, F002, F004, F008); defines phenotype spectrum, isolated nature, consanguineous AR inheritance, and LAR molecular function.
9245518	*Impaired mammary gland development and function in mice lacking LAR*	Model organism (mouse KO)	Corroborates essential mammary role of *Ptprf*/LAR (F003); demonstrates lactation defect; provides protein-domain architecture. Caveat: mouse defect is functional/lactational, not structural athelia.
30550377	*Hypogonadotropic hypogonadism… in Ulnar-Mammary syndrome*	Human clinical	Defines UMS (*TBX3*) as a syndromic differential (F005).
32067224	*Novel phenotype… TP63 molecular defect*	Human clinical	Defines TP63 limb-mammary syndrome differential with limb/ectodermal features absent in BNAH2 (F005).
32083401	*Phenotypic expansion of KMT2D-related disorder*	Human clinical	Describes KMT2D-related athelia with choanal atresia/hypoparathyroidism — key mimic distinguished from BNAH2 (F005).
38281558	*Spectrum of TP63-related disorders*	Human clinical	Catalogs TP63 ectodermal-dysplasia features (hypoplastic breasts/nipples) distinguishing from isolated BNAH2 (F005).
39320041	*Inherited TBX3 alteration in ulnar-mammary syndrome*	Human + Drosophila model	Details UMS core features (ulnar defects, hypoplastic nipples/areolas) for differential diagnosis (F005).
23896162	*Bilateral congenital amazia: systematic review*	Clinical systematic review	Quantifies reconstruction modalities — mainstay of management (F006).

PMID	Title (abbrev.)	Evidence type	How it supports / challenges the findings
33830564	*Breast reconstruction via fat grafting (Finlay-Marks)*	Case report	Documents fat grafting as a reconstructive option (F006).
31585491	*P63-related disorders: dermatological characteristics*	Human clinical	Supports TP63 ectodermal differential (F005).
33930582	*TBX3 and TBX5 duplication: Holt-Oram/ UMS overlap*	Human genetics	Supplementary TBX3/UMS differential context (F005).
32016944	*Poland Syndrome with Xp22.31 duplication*	Human case	Poland syndrome differential (breast/ nipple + pectoral anomalies).

Additional literature reviewed during triage — tangential to BNAH2 — included the Bork-Baykal nipple-areola sparing phenomenon (PMID: 30989705) and the McGovern-nipple airway device in CHARGE syndrome (PMID: 32611869).

Section-by-Section Synthesis

1. Disease Information

BNAH2 is an isolated congenital disorder of nipple-areola and breast development. **Identifiers:** MONDO:0014450; OMIM:616001; GARD:0016044; MedGen/UMLS C4014918 (CUI 863355). No dedicated Orphanet or specific ICD-10/ICD-11 code is established; it falls under broad categories for congenital breast/nipple malformation. **Synonyms:** "Breasts and/or nipples, aplasia or hypoplasia of, 2"; athelia (isolated); isolated congenital athelia/hypothelia. Information is derived from an **aggregated disease-level resource** basis (OMIM/MONDO) built on a single primary clinical report, not from EHR/patient-level data.

2. Etiology

Causal factor: genetic — biallelic (homozygous) loss-of-function in *PTPRF*. **Genetic risk factor:** homozygosity for a *PTPRF* LoF allele, enabled by consanguinity (founder effect). **Environmental risk factors:** none identified; **consanguinity** is the principal demographic risk driver. **Protective factors:** none specifically identified; in practical terms, outbreeding reduces recessive risk. **Gene-environment interactions:** none documented — the phenotype appears determined by genotype with variable expressivity.

3. Phenotypes

Congenital, non-progressive spectrum: athelia (HP:0002561), hypoplastic nipples (HP:0002557), aplasia/hypoplasia of nipples (HP:0006709), hypoplastic areola (HP:0100853), breast aplasia/hypoplasia (HP:0100783, HP:0010311, HP:0003187), uni-/bilateral breast hypoplasia (HP:0012813/HP:0012814), and abnormal eyebrow morphology (HP:0000534). Onset neonatal/congenital; severity variable (mild unilateral to severe bilateral); frequency within the single reported family: high penetrance among homozygotes. **QoL impact:** cosmetic/body-image and psychosocial effects; inability to breastfeed in affected females; no impact on general health or lifespan.

4. Genetic/Molecular Information

Causal gene: *PTPRF* (HGNC:9670; gene MIM 179590; Entrez 5792; UniProt P10586), chromosome 1p34.2. **Variant:** homozygous truncating frameshift (loss-of-function). **Classification:** pathogenic (biallelic, segregating, LoF in a constrained gene). **Allele frequency:** biallelic LoF essentially absent from gnomAD; carrier frequency extremely low. **Origin:** germline. **Functional consequence:** loss of function. **Modifier genes/epigenetics/chromosomal abnormalities:** none reported.

5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. BNAH2 is a monogenic developmental disorder.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** above for the ordered causal chain. Core pathway elements: adherens-junction–associated LAR tyrosine phosphatase activity, canonical Wnt signaling, and peptide growth-factor signaling governing ectodermal mammary-ridge morphogenesis.

7. Anatomical Structures Affected

Primary: nipple (UBERON:0002030), areola (UBERON:0002032), breast (UBERON:0000310), mammary gland (UBERON:0001911). **System:** integumentary/reproductive (mammary). **Tissue:** ectoderm-derived epithelium. **Cells:** mammary epithelial populations (CL:0002327, CL:0002326, CL:0002324, CL:0002325), keratinocytes (CL:0000312). **Subcellular:** plasma membrane / adherens junction (GO:0005886). **Lateralization:** bilateral, unilateral, or asymmetric.

8. Temporal Development

Onset: congenital. **Course:** stable, non-progressive, lifelong structural anomaly. No disease stages, remission, or episodic pattern. **Critical period:** embryonic mammary-ridge development (window for any theoretical developmental intervention).

9. Inheritance and Population

Inheritance: autosomal recessive. **Epidemiology:** ultra-rare (single reported family; no prevalence/incidence figures). **Penetrance:** high in homozygotes; **expressivity:** variable. **Consanguinity:** central. **Founder effect:** shared ancestral allele in an Israeli Arab kindred. **Sex ratio:** both sexes affected (no evident skew). **Carrier frequency:** extremely low in the general population.

10. Diagnostics

Clinical: physical examination identifying athelia/hypothelia and breast hypoplasia from birth; assessment for associated anomalies to exclude syndromes. **Genetic testing:** molecular confirmation by single-gene *PTPRF* sequencing, gene panels for congenital breast/nipple anomaly, or WES/WGS; homozygosity mapping is informative in consanguineous pedigrees. **Imaging:** ultrasound/MRI to characterize residual breast parenchyma before reconstruction. **Differential diagnosis:** UMS (*TBX3*), TP63-related disorders, scalp-ear-nipple syndrome (*KCTD1*), KMT2D-related disorder, Poland syndrome — distinguished by their extramammary features.

11. Outcome / Prognosis

Excellent — normal lifespan, no systemic morbidity or mortality. Morbidity is limited to cosmetic/psychosocial impact and (in females) inability to lactate. **Prognostic factors:** none affecting survival; extent of breast tissue influences reconstructive outcome.

12. Treatment

No pharmacologic or disease-modifying therapy. **Reconstructive surgery** (prosthetic implant, autologous tissue, autologous + prosthesis, or fat grafting) and **nipple-areola reconstruction**, generally deferred to breast maturity/adolescence; mean age at reconstruction ~27.6 years in congenital breast anomaly cohorts. **Supportive:** psychological support, lactation counseling where relevant. NCIT concepts: breast reconstruction, breast implant, autologous fat transfer.

13. Prevention

No primary medical prevention. **Genetic counseling** for consanguineous families, **carrier/cascade testing**, and **prenatal or preimplantation genetic diagnosis** are the available preventive measures. No immunization or environmental intervention applies.

14. Other Species / Natural Disease

Ptprf is conserved across mammals and vertebrates (orthologs listed in Finding 7). No naturally occurring companion-animal or wildlife disease specifically attributed to *PTPRF* nipple/breast aplasia is documented in the reviewed literature; the mouse knockout is an engineered model, not a natural disease. No zoonotic relevance.

15. Model Organisms

Mouse (*Mus musculus*, NCBI:txid10090): LAR/*Ptprf* phosphatase-domain knockout (Schaapveld et al., PMID: 9245518). **Phenotype recapitulation:** partial — the model reproduces an essential mammary role (impaired alveolar terminal differentiation, lactation failure) but manifests a **functional** defect rather than the **structural** athelia seen in humans. **Limitation:** overall normal development and histology in $LAR^{-/-}$ mice indicates species-specific redundancy for nipple morphogenesis. **Applications:** studying LAR's role in mammary differentiation, adhesion, and Wnt/growth-factor signaling. Resources: MGI (mouse *Ptprf*, Entrez 19268); orthologs available in RGD, ZFIN (zebrafish 140819), etc.

Limitations and Knowledge Gaps

1. **Single-family evidence.** The entire human genetic basis of BNAH2 rests on **one consanguineous family with three affected individuals** (PMID: 24781087). No

independent replication or additional *PTPRF* families have been reported, so penetrance, full expressivity range, and genotype–phenotype correlations remain uncertain.

2. **Mechanism largely inferred.** The adherens-junction/Wnt/growth-factor signaling steps are biologically plausible and supported by LAR's known molecular functions, but the specific developmental mechanism in human nipple/mammary-ridge formation has **not been experimentally demonstrated**.
3. **Human–mouse phenotypic discordance.** The *Ptprf*-null mouse shows a functional lactation defect with otherwise normal development, not structural athelia — indicating species-specific differences and limiting direct mechanistic extrapolation.
4. **No epidemiologic data.** Prevalence, incidence, carrier frequency in specific populations, and any geographic/variant-specific distribution beyond the index kindred are unknown.
5. **No natural-history or QoL studies.** Formal quality-of-life, psychosocial-outcome, and reconstructive-outcome data **specific to BNAH2** do not exist; management evidence is extrapolated from broader congenital breast anomaly cohorts.
6. **Ontology/identifier gaps.** No dedicated Orphanet number or specific ICD-11 code was confirmed for this entity.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment and replication.** Query GeneMatcher, DECIPHER, and large consanguineous-population cohorts (e.g., Middle Eastern, South Asian biobanks) for additional biallelic *PTPRF* LoF individuals to replicate the gene–disease association and refine the phenotype.
2. **Functional validation of the developmental mechanism.** Use human mammary organoids / iPSC-derived mammary models with *PTPRF* knockout to test whether loss of LAR impairs nipple-placode/mammary-bud morphogenesis via adherens-junction and Wnt readouts; perform rescue experiments.
3. **Improved animal modeling.** Generate conditional/ectoderm-restricted *Ptprf* models or test redundancy with related type IIa RPTPs (*PTPRD*, *PTPRS*) to determine why mice escape structural athelia; consider zebrafish or other models of nipple-equivalent/mammary-line development.

4. **Molecular profiling.** Transcriptomic (GEO/single-cell) and proteomic characterization of *PTPRF*-null mammary epithelium to map downstream Wnt and growth-factor signaling perturbations and identify LAR substrates (e.g., EPHA2, β-catenin pathway components).
5. **Clinical/ontology curation.** Pursue an Orphanet entry and ICD-11 extension code; formalize the HPO/UBERON/CL/GO annotation set (above) in the knowledge base; register the gene–disease relationship in ClinGen for formal validity classification.
6. **Patient-centered outcomes.** Establish a registry (even for ultra-rare N) capturing reconstructive timing/modalities, lactation outcomes in affected females, and psychosocial QoL using validated instruments (e.g., BREAST-Q), to generate BNAH2-specific management evidence.

Report compiled from 5 discovery iterations, 9 confirmed findings, and 14 reviewed papers. Evidence source types: human clinical genetics (PMID: 24781087); mouse model (PMID: 9245518); clinical review (PMID: 23896162); differential-diagnosis clinical literature (TBX3, TP63, KMT2D, KCTD1); plus ontology/constraint database verification (MONDO, OMIM, gnomAD, OLS/HP/CL/UBERON/GO).
